When and Why Structure? Static and Dynamic Decision Problems

Empirical Methods — Week 6

Teaching deck draft

Week 6 question

- When does an applied question require an economic model rather than only a credible comparison?
- What is observed, what is latent, and what variation disciplines the latent objects?
- When do expectations, dynamics, welfare, or equilibrium response change the research design?
- How should a reader interpret a structural counterfactual?

Structure is a research choice, not a rival ideology to design-based work.

Why structure?

- Latent primitives: preferences, costs, technologies, beliefs, skills, frictions.
- Expectations: decisions depend on future wages, offers, policy rules, or risks.
- Dynamic incentives: actions today change tomorrow's state.
- Counterfactuals: the policy of interest may be outside observed support.
- Welfare: utility, equivalent variation, and insurance value are not directly observed.
- Equilibrium response: prices, queues, vacancies, and market composition may adjust.

The model should buy an object that the data cannot reveal transparently without structure.

Static vs dynamic problems

Type	Example	Key latent object	Main risk
Static choice	job or school menu	contemporaneous utility tradeoffs	wrong menu or shock structure
Static labor supply	hours under tax schedule	preferences over consumption and leisure	functional-form dependence
Dynamic schooling	school, work, occupation	continuation values and skill accumulation	state-space misspecification
Dynamic search	accept offer or search	offer distribution and reservation value	expectations and duration dependence
Lifecycle labor supply	work, saving, retirement	risk, insurance, intertemporal preferences	extrapolation and heterogeneity

Observed and latent objects

Observed in data	Latent in model	Identification question
choices and transitions	preferences and switching costs	which choices shift when states or prices change?
wages and employment	skill prices and productivity	which histories discipline returns and shocks?
policy rules and prices	perceived incentives and beliefs	do agents know and respond to the rule?
state histories	transition laws and option values	which state changes predict future behavior?
outcomes	welfare and compensating variation	how does utility map to observed behavior?

A structural paper should make this map visible before the main counterfactual.

Static latent utility

$$U_{ij} = u_j(X_i, Z_{ij}; \theta) + \alpha_i q_{ij} + \xi_j + \varepsilon_{ij}$$

$$d_{ij} = \mathbf{1}\{U_{ij} \geq U_{ik} \text{ for all } k \in \mathcal{J}_i\}$$

$$\mathbb{P}(d_{ij} = 1 \mid X_i, Z_i; \theta) = \frac{\exp(V_{ij})}{\sum_{k \in \mathcal{J}_i} \exp(V_{ik})}$$

- Observed: choice, menu, attributes, prices or wages.
- Latent: utility, nonpecuniary value, unobserved quality, shocks.
- The shock assumption turns latent utility into choice probabilities.

Dynamic Bellman logic

$$V(s_t) = \max_{a_t \in \mathcal{A}(s_t)} \{u(a_t, s_t; \theta) + \delta \mathbb{E}_\theta [V(s_{t+1}) \mid s_t, a_t]\}$$

$$v(a, s; \theta) = u(a, s; \theta) + \delta \sum_{s'} V(s'; \theta) F(s' \mid s, a; \theta)$$

- States summarize payoff-relevant history.
- Actions change future states and future options.
- Continuation values are the object reduced-form comparisons often leave latent.

Continue vs stop

$$v_C(s;\theta) = u_C(s;\theta) + \delta \sum_{s'} V(s';\theta) F_C(s' \mid s;\theta)$$

$$v_S(s;\theta) = u_S(s;\theta) + \delta \sum_{s'} V(s';\theta) F_S(s' \mid s;\theta)$$

$$\mathbb{P}(a = S \mid s;\theta) = \frac{\exp(v_S(s;\theta))}{\exp(v_S(s;\theta)) + \exp(v_C(s;\theta))}$$

- Replacement: keep operating or replace an engine.
- Search: accept a job or continue searching.
- Schooling: stay in school or work now.

Estimation strategies

Likelihood

$$\widehat{\theta} = \arg\max_{\theta} \sum_{i,t} \log \mathbb{P}_{\theta}(a_{it} \mid s_{it}, p_{it}) + \log f_{\theta}(y_{it}, s_{i,t+1} \mid a_{it}, s_{it}, p_{it})$$

Moments or simulation

$$\widehat{\theta} = \arg\min_{\theta} \left[\widehat{m}^{data} - m^{model}(\theta)\right]' W \left[\widehat{m}^{data} - m^{model}(\theta)\right]$$

- Fit should report targeted moments and non-target validation.
- Identification asks which variation disciplines each primitive.

Policy counterfactuals and welfare

$$\widehat{\theta} \Longrightarrow \{\sigma_{\widehat{\theta}}(a \mid s; p), F_{\widehat{\theta}}(s' \mid s, a; p), Y_{\widehat{\theta}}(p), W_{\widehat{\theta}}(p)\}$$

$$\Delta Y(p_1, p_0) = \mathbb{E}_{\widehat{\theta}}[Y_i(p_1) - Y_i(p_0)], \qquad \Delta W(p_1, p_0) = \mathbb{E}_{\widehat{\theta}}[W_i(p_1) - W_i(p_0)]$$

- The policy changes rules, prices, constraints, or menus.
- The model holds some primitives fixed and recomputes behavior.
- Credibility falls as the counterfactual moves away from identifying support.

Theory to applied papers

Paper	Question	What is latent?	Model contribution
Rust	replace now or later?	replacement costs and value of waiting	dynamic replacement counterfactuals
Hotz-Miller	estimate dynamic choice?	value differences	CCP-based estimation logic
Keane-Wolpin	schooling, work, career?	skills, preferences, expectations	dynamic career paths
Todd-Wolpin	schooling subsidy?	household behavioral rules	validation and policy simulation
Low-Meghir-Pistaferri	lifecycle risk?	insurance and risk processes	welfare under wage and employment risk
Adda-Dustmann-Stevens	child career costs?	dynamic timing and persistence	long-run career counterfactuals

Credibility threats

- Functional form: results driven by parametric utility, shocks, or transitions.
- Expectations: agents may not know or believe the process assumed by the model.
- Latent heterogeneity: flexible types can absorb misspecification.
- Off-target fit: the model matches targeted moments but misses relevant behavior.
- Identification by assumption: key primitives come from normalization more than data.
- Extrapolation: the counterfactual changes support, menus, or equilibrium conditions.

Structural discipline means reporting these limits, not hiding them.

Research Lab design

Reproduce

- Rust-style replacement
- Mileage states
- Replacement CCPs
- Dynamic logit likelihood

Diagnose

- Static vs dynamic fit
- State support
- Parameter sensitivity
- Replacement-cost counterfactual

Transfer

- Career-choice panel
- School/work states
- Tuition subsidy
- Counterfactual memo

Takeaways

- Use structure when the research object is latent, dynamic, welfare-relevant, or outside observed support.
- The model must identify what is observed, what is latent, and what data variation disciplines the mapping.
- Bellman logic is the language of dynamic incentives and continuation values.
- Estimation is credible only when fit, validation, and sensitivity are reported transparently.
- Counterfactuals should be interpreted through the distance between the new policy and the evidence that identified the primitives.